

Random cardinalities of coalescence classes in a six-state Markov grand coupling

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Abstract

We construct an independent random map grand coupling of a strictly positive six-state transition matrix for which the eventual coalescence classes have cardinalities $(2, 4)$ or $(3, 3)$, each with probability $1/2$. The number of classes is always two, giving a negative answer to Open Problem 3.11 of Grimmett and Holmes, *Coalescence in Markov Chains* (2026). The transition matrix has identical rows and satisfies $P^2 = P$. A pathwise proof uses two proper colorings of the disjoint union of two three-vertex paths. We extend the construction to bipartite graphs, characterize when that construction has a random cardinality profile, and derive a second counterexample by splitting states in Grimmett and Holmes' four-state example. A self-contained proof of the classical equal stationary mass theorem explains why cardinalities can vary while stationary masses remain equal, and shows that six states are minimal among finite irreducible independent random map couplings. The same mass constraint gives an affirmative answer to Open Problem 3.8: Almost every strictly positive transition matrix admits only couplings with one eventual class. For matrices with identical rows, we characterize all possible numbers of classes by partitions of equal stationary mass.

Keywords: Markov chains; grand couplings; coalescence; random mappings; transformation semigroups.

1. The question and the result

Let S be a nonempty finite set, and let μ be a probability measure on the set S^S of functions from S to itself. Let $F_1, F_2, \dots$ be independent with common law μ . We use ordinary function composition, with the rightmost function applied first, and put

$$H_0 = \text{id}_S, \quad H_t = F_t \circ \dots \circ F_1, \quad X_t^i = H_t(i). \quad (1)$$

The same function F_t updates every coordinate at time t . For each fixed i , the process X^i is a Markov chain with transition probabilities

$$P_{xy} = \mu\{f : f(x) = y\}, \quad x, y \in S. \quad (2)$$

Indeed, F_{t+1} is independent of the maps already used, so its conditional action on the current state x has the distribution in (2). We assume throughout, except where explicitly stated otherwise, that P is irreducible.

If two coordinates meet, they subsequently coincide, since future functions act identically on equal inputs. Thus

$$i \sim j \iff H_t(i) = H_t(j) \text{ for some } t \geq 0$$

is an equivalence relation. Write $\mathcal{C}_\infty$ for its random partition of S . For a partition $\mathcal{C}$, define $\text{prof}(\mathcal{C})$ to be the tuple of its block cardinalities arranged in nondecreasing order, with repetitions retained. This tuple records the multiset of cardinalities without choosing labels for the blocks.

Grimmett and Holmes [3, Open Problem 3.11] ask whether these cardinalities must be deterministic. The relevant distinction is between the number of eventual classes and their sizes. The former is almost surely constant; an elementary minimum rank proof is recalled in Section 4. The following construction shows that the latter need not be.

Theorem 1.1. *There is a probability measure μ supported on sixteen maps of $S = \{1, 2, 3, 4, 5, 6\}$, each of probability $1/16$, consistent with*

$$P = \mathbf{1}\pi, \quad \pi = \frac{1}{8}(1, 2, 1, 1, 2, 1), \quad (3)$$

where $\mathbf{1}$ is the column vector of six ones, such that

$$\mathbb{P}\{\text{prof}(\mathcal{C}_\infty) = (2, 4)\} = \mathbb{P}\{\text{prof}(\mathcal{C}_\infty) = (3, 3)\} = \frac{1}{2}. \quad (4)$$

For every sequence of maps from this support, the eventual partition is already the partition into fibers of F_1 . In particular, there are exactly two classes on every such sequence. The matrix P is strictly positive, irreducible and aperiodic, and $P^2 = P$.

Earlier examples show that the eventual partition can be random even when its cardinality profile is fixed. In the four-state coupling of Grimmett and Holmes [3, Example 3.10], the two possible partitions both have profile $(2, 2)$, and the two minimum rank partitions in the six-state example of Budzban and Feinsilver [1, Sections 3–4] likewise share the profile $(2, 2, 2)$. The construction above gives different cardinality profiles under a single coupling law. Corollary 7.2 shows more generally how a finite state-splitting lift converts a nonconstant eventual partition into a nonconstant cardinality profile.

The equal stationary mass theorem also answers two questions about the possible number of classes. Theorem 5.1 and Corollary 5.2 give an affirmative answer to Open Problem 3.8 of [3]. Theorem 5.3 determines the possible numbers for every strictly positive matrix with identical rows. In particular, when all transitions have probability $1/n$, the possible numbers are exactly the divisors of n . These are applications of the classical mass constraint; the construction for identical rows is closely related to the work of Gusev and Pribavkina [4].

The main construction and its proof appear in Section 2. Section 3 gives a graph formulation and an infinite family. Section 4 proves the equal stationary mass theorem and the minimality of six states. Section 5 gives the additional results on possible numbers of classes. Sections 6 and 7 describe the semigroup structure and derive another counterexample directly from a previously published coupling.

2. The six-state construction

Consider the simple undirected graph G with vertex set S and edge set

$$E = \{\{1, 2\}, \{2, 3\}, \{4, 5\}, \{5, 6\}\}. \quad (5)$$

It is the disjoint union of two paths on three vertices. Introduce two subsets

$$A_0 = \{1, 3, 4, 6\}, \quad A_1 = \{1, 3, 5\}. \quad (6)$$

For either $b \in \{0, 1\}$, every edge has one endpoint in A_b and one in $S \setminus A_b$, i.e., $(A_b, S \setminus A_b)$ is a proper two-coloring.

Let $\vec{E}$ be the eight oriented edges obtained by giving each edge in (5) both orientations.



Figure 1: The two colorings used in the construction. Shaded vertices form A_b . Reversing the colors on the second connected component changes the cardinalities but preserves the condition that adjacent vertices have different colors..

Choose B uniformly from $\{0, 1\}$ and, independently, choose (U, V) uniformly from $\vec{E}$. Define

$$f_{b,u,v}(x) = \begin{cases} u, & x \in A_b, \\ v, & x \notin A_b. \end{cases} \quad (7)$$

The law μ is the law of $f_{B,U,V}$. The choices give sixteen distinct maps: the two values of b have different fiber partitions, and, for fixed b , the values on the two parts recover the ordered pair (u, v) . Equivalently, in the notation $(f(1), \dots, f(6))$, they have the forms

$$f_{0,u,v} = (u, v, u, u, v, u), \quad f_{1,u,v} = (u, v, u, v, u, v). \quad (8)$$

cf. the Appendix for all sixteen maps.

2.1 Verification of the single-coordinate transition matrix

The degrees of the six vertices of G are $(1, 2, 1, 1, 2, 1)$. Since every oriented edge is equally likely,

$$\mathbb{P}(U = j) = \mathbb{P}(V = j) = \frac{\deg_G(j)}{8} = \pi_j. \quad (9)$$

For fixed x and fixed b , the output $f_{b,U,V}(x)$ is either U or V , according to the membership of x in A_b . Both endpoint distributions are π , so

$$\mathbb{P}(F_t(x) = j \mid B_t = b) = \pi_j.$$

Averaging over b proves (3). In particular, every transition has positive probability and every state has a positive self-transition. Furthermore,

$$\pi P = (\pi \mathbf{1})\pi = \pi, \quad P^2 = \mathbf{1}(\pi \mathbf{1})\pi = \mathbf{1}\pi = P.$$

Each individual coordinate has distribution π after one step, and its states at times $1, 2, \dots$ are independent with common law π . This assertion concerns one coordinate; the coordinates updated by the same random maps are dependent.

2.2 The pathwise argument

Lemma 2.1. *Every map in (7) sends the two endpoints of every edge of G to the two distinct endpoints of an edge of G .*

Proof. If $\{x, y\} \in E$, then exactly one of x, y belongs to A_b . Consequently,

$$\{f_{b,u,v}(x), f_{b,u,v}(y)\} = \{u, v\} \in E.$$

Since the graph has no loops, then $u \neq v$. $\square$

Proof of Theorem 1.1. The matrix assertions were established above. Fix any infinite sequence of allowed maps. The first map has image $\{u_1, v_1\}$, an edge of G , and fibers A_{b_1} and $S \setminus A_{b_1}$. All starting states in either fiber therefore meet at time one.

At time one the two remaining locations are the endpoints of an edge. Lemma 2.1 implies that, after the next map, they are still the endpoints of an edge and remain distinct. Induction gives the same statement at every subsequent time. Hence states in different fibers of the first map never meet, while states in the same fiber have already met, so that

$$\mathcal{C}_\infty = \{A_{B_1}, S \setminus A_{B_1}\}. \quad (10)$$

The cardinalities of A_0 and its complement are 4 and 2; those of A_1 and its complement are 3 and 3. Since B_1 is fair, (4) follows. $\square$

Remark 2.2 (The profile law can vary while P stays fixed). Replace the fair choice of B by $\mathbb{P}(B = 0) = \theta$, where $0 \leq \theta \leq 1$, and keep the independent uniform oriented edge choice. Equation (9) still gives the same matrix P . The probabilities of the profiles (2, 4) and (3, 3) are then θ and $1 - \theta$. In contrast, choosing the constant map $x \mapsto J$ with $J \sim \pi$ also gives the same P , but all six starting states meet at time one. Thus even this particularly simple transition matrix supports markedly different joint behavior.

3. A general construction from bipartite graphs

Let $G = (S, E)$ be a finite simple bipartite graph with at least one edge and no isolated vertices. Write $m = |E|$. A subset $A \subset S$ will be called a color class if every edge has exactly one endpoint in A . Let ρ be any probability law on such subsets. Choose $A \sim \rho$ independently of a uniform oriented edge (U, V) and define $f_{A,U,V}$ by (7), with A in place of A_b .

Proposition 3.1. *The resulting independent random map coupling has*

$$P_{ij} = \pi_j = \frac{\deg_G(j)}{2m} > 0. \quad (11)$$

For every allowed sequence of maps, its eventual partition is $\{A^{(1)}, S \setminus A^{(1)}\}$, where $A^{(1)}$ is the first chosen color class. Each of these two blocks has stationary mass $1/2$.

Proof. Both coordinates of a uniform oriented edge have distribution $\deg_G(j)/(2m)$, proving (11). Every allowed map sends edges to edges, so the proof of (10) applies without change. Finally, each edge has precisely one endpoint in A , whence

$$\sum_{j \in A} \deg_G(j) = m, \quad \pi(A) = \frac{m}{2m} = \frac{1}{2}.$$

$\square$

Note that the auxiliary graph may be disconnected even though P is strictly positive, since the maps may send a color class to an endpoint in any connected component.

3.1 Exactly when this graph construction has varying sizes

Suppose G has connected components $G_1, \dots, G_k$. Let their bipartition sizes be (a_i, b_i) , relabeled so that $a_i \geq b_i$, and put

$$n = |S|, \quad \delta_i = a_i - b_i \geq 0.$$

Since each component is connected and has an edge, its proper two-coloring is unique up to exchange of colors. Thus the possible cardinalities of a color class satisfy

$$|A| = \frac{1}{2} \left(n + \sum_{i=1}^k \varepsilon_i \delta_i \right), \quad \varepsilon_i \in \{-1, 1\}. \quad (12)$$

Its unordered profile depends only on the absolute value of the signed sum:

$$\text{prof}\{A, S \setminus A\} = \left(\frac{n - |\sum_i \varepsilon_i \delta_i|}{2}, \frac{n + |\sum_i \varepsilon_i \delta_i|}{2} \right).$$

Proposition 3.2. *Assume ρ gives positive probability to every proper two-coloring. The profile in Proposition 3.1 is nonconstant if and only if at least two connected components have unequal bipartition sizes. Consequently, six vertices are necessary and sufficient for a counterexample within this graph construction.*

Proof. If at most one δ_i is positive, then $|\sum_i \varepsilon_i \delta_i|$ is independent of the signs. If at least two are positive, let $D = \sum_i \delta_i$ and choose j with $0 < \delta_j < D$. All positive signs give absolute sum D , and reversing only the j th sign gives $|D - 2\delta_j| < D$. Both choices have positive probability and yield different profiles.

A connected bipartite graph with an edge and unequal bipartition sizes has at least three vertices. Two such components require at least six vertices. The graph in (5) attains that bound. Proposition 4.4 below proves the same lower bound for all finite irreducible independent random map couplings by a different argument. $\square$

For an explicit infinite family, let G be the disjoint union of two stars $K_{1,d}$, with $d \geq 2$. Choose with equal probabilities between putting both centers in one color class and putting one center together with the leaves of the other star. Proposition 3.1 gives profiles

$$(2, 2d) \quad \text{and} \quad (d+1, d+1), \quad (13)$$

each with probability $1/2$. The two centers each have stationary mass $1/4$; each of the $2d$ leaves has mass $1/(4d)$. In either profile, each block again has mass $1/2$.

4. Why the stationary masses are equal

The equality of masses in Proposition 3.1 reflects a classical fact about finite random mappings. Equal-weight partitions occur in Friedman's road coloring analysis [2, Theorem 2.1]. The minimum rank fiber statement appears in Yano [6, Theorem 4.1] and in Budzban and Feinsilver [1, Theorem 4.8]. We give a self-contained proof, including the passage from minimum rank maps to eventual classes. The theorem numbers in these two references refer to the linked preprints. Only irreducibility is needed for the version below.

Let $\mathcal{A} = \text{supp } \mu$ and let $\mathcal{T}$ be the finite semigroup of nonempty compositions of maps from $\mathcal{A}$. For $g \in \mathcal{T}$, define its rank to be $|g(S)|$, and define

$$r = \min_{g \in \mathcal{T}} |g(S)|. \quad (14)$$

A *fiber* of h is a nonempty set $h^{-1}\{z\}$, with $z \in h(S)$. A subset is a *transversal* of a partition if it contains exactly one point from every block.

Lemma 4.1 (Transversals of minimum rank fibers). *If $g, h \in \mathcal{T}$ both have rank r , then $g(S)$ is a transversal of the fiber partition of h . Moreover, every $f \in \mathcal{T}$ acts injectively on $g(S)$.*

Proof. The composition $h \circ g$ has rank at least r by minimality and at most r because $g(S)$ has r elements. Therefore h is injective on $g(S)$. Its r distinct values must be all of $h(S)$, proving the transversal assertion. The same argument applied to $f \circ g$ proves the final assertion. $\square$

Theorem 4.2 (Equal stationary masses). *Let P in (2) be irreducible, with stationary distribution π , and let r be as in (14). Every fiber C of every rank- r map in $\mathcal{T}$ satisfies*

$$\pi(C) = \frac{1}{r}. \quad (15)$$

Almost surely, $\mathcal{C}_\infty$ is the fiber partition of a rank- r map in $\mathcal{T}$. Consequently it has r blocks, each of stationary mass $1/r$.

Proof. Fix a rank- r map h and one of its fibers C . Also fix any rank- r image $R = g(S)$. For each $t \geq 0$, let $R_t = H_t(R)$ and assign mass $1/r$ to each of its points. By Lemma 4.1, H_t is injective on R and R_t is another rank- r image. Every R_t therefore meets C in exactly one point. Pathwise,

$$\frac{1}{r} \sum_{x \in R} \mathbf{1}_{\{H_t(x) \in C\}} = \frac{1}{r}. \quad (16)$$

Let ν_R be the uniform probability measure on R . Taking expectations in (16) and using the marginal Markov chains gives

$$(\nu_R P^t)(C) = \frac{1}{r} \quad (t \geq 0). \quad (17)$$

For completeness, the Cesàro averages $\bar{\nu}_N = N^{-1} \sum_{t=0}^{N-1} \nu_R P^t$ converge to π . Indeed, the probability simplex is compact, and

$$\bar{\nu}_N P - \bar{\nu}_N = \frac{\nu_R P^N - \nu_R}{N} \rightarrow 0.$$

Every subsequential limit is thus stationary, and irreducibility gives the unique stationary law π . Applying this to (17) proves (15). This argument also covers periodic P .

It remains to identify the eventual partition. Choose a word $w = f_\ell \circ \dots \circ f_1$ of rank r , with each $f_j \in \mathcal{A}$. In each disjoint block of ℓ independent draws, the ordered word $(f_1, \dots, f_\ell)$ has probability

$$p = \prod_{j=1}^{\ell} \mu(f_j) > 0.$$

The probability that it fails to occur in the first N such blocks is $(1-p)^N$, which tends to zero. At the end of its first occurrence, the full composition is $w \circ H$ for the preceding composition H . Its image has at most r points and by (14), exactly r points.

Thereafter the rank cannot decrease below r . Fiber partitions can only merge under further left composition; with their number fixed at r , no merger is possible. Thus the partition at this finite time is exactly $\mathcal{C}_\infty$. The already proved equality applies to all its blocks. $\square$

Corollary 4.3. *If π is uniform on n states, then every grand coupling has deterministic eventual profile*

$$\text{prof}(\mathcal{C}_\infty) = (n/r, \dots, n/r)$$

with r entries, where r is its minimum rank. In particular r divides n . A random eventual profile for an irreducible chain requires a nonuniform stationary distribution.

Proof. For every eventual block C , Theorem 4.2 gives $|C|/n = \pi(C) = 1/r$. $\square$

Proposition 4.4 (Six states are minimal). *An independent random map grand coupling of an irreducible chain on at most five states has an almost surely deterministic eventual*

cardinality profile. Consequently, Theorem 1.1 has the smallest possible state space size under these assumptions.

Proof. Fix a coupling, and let r be its minimum rank. By Theorem 4.2, every eventual block has mass $1/r$. Since irreducibility gives $\pi_x > 0$ for every state, a state x is a singleton block if and only if $\pi_x = 1/r$. The set of singleton blocks is therefore deterministic.

Let k be the number of these singleton states and let $q = r - k$ be the number of remaining blocks. Both numbers are fixed. Each remaining block has size at least two. If $q = 0$, all blocks are singletons. If $q = 1$, the sole remaining block has the fixed size $n - k$. If $n \leq 5$, the only other possibility is $q = 2$, and its two sizes sum to $n - k \leq 5$. Two integers each at least two with fixed sum at most five have the unique unordered pair $(2, 2)$ or $(2, 3)$. Thus the profile cannot vary. The six-state construction supplies the matching upper bound. $\square$

In the six-state example the masses can also be checked immediately:

$$\begin{aligned}\pi(A_0) &= \pi(\{1, 3, 4, 6\}) = 4 \cdot \frac{1}{8} = \frac{1}{2}, & \pi(S \setminus A_0) &= \pi(\{2, 5\}) = 2 \cdot \frac{1}{4} = \frac{1}{2}, \\ \pi(A_1) &= \pi(\{1, 3, 5\}) = \frac{1}{8} + \frac{1}{8} + \frac{1}{4} = \frac{1}{2}, & \pi(S \setminus A_1) &= \pi(\{2, 4, 6\}) = \frac{1}{4} + \frac{1}{8} + \frac{1}{8} = \frac{1}{2}.\end{aligned}$$

Thus the random cardinalities are compatible with a rigid deterministic mass constraint. Equal stationary mass does not imply equal counting measure when the states have different stationary weights. The number of classes is two in either case, and each class has stationary mass $1/2$. Their cardinalities nevertheless have profiles $(2, 4)$ and $(3, 3)$: Four states of mass $1/8$ balance two states of mass $1/4$, while a different grouping gives three states on each side, thus illustrating how unequal stationary weights permit variation in cardinalities. Corollary 4.3 shows that such variation is impossible when the stationary distribution is uniform.

5. Possible numbers of coalescence classes

For an irreducible transition matrix P on $S = \{1, \dots, n\}$, write $k(\mu)$ for the almost surely constant number of eventual classes under a law μ consistent with P . By Theorem 4.2, this is the minimum rank of the semigroup generated by $\text{supp } \mu$. Following [3], put

$$K(P) = \{k(\mu) : \mu \text{ is consistent with } P\}. \quad (18)$$

This set is nonempty: choosing the values $F(x)$ independently, with $\mathbb{P}(F(x) = y) = P_{xy}$, always gives a consistent law. In this section, π_P denotes the stationary distribution of P when the matrix itself varies.

Theorem 4.2 imposes a restriction on every member of $K(P)$. If $r \in K(P)$, then S admits a partition into r nonempty sets, each of π_P -mass $1/r$. In particular,

$$K(P) \neq \{1\} \implies \pi_P(C) = \frac{1}{r} \text{ for some } \emptyset \neq C \subsetneq S, \quad 2 \leq r \leq n. \quad (19)$$

Although there are infinitely many consistent laws, there are only finitely many pairs (C, r) in this condition. This finiteness allows us to control all consistent laws at once.

5.1 Coalescence for almost every transition matrix

Let $\mathcal{P}_n^+$ be the set of strictly positive $n \times n$ stochastic matrices. For $n \geq 2$, we give this set coordinates by retaining the first $n - 1$ entries of each row and setting

$$P_{in} = 1 - \sum_{j=1}^{n-1} P_{ij}$$

so that $\mathcal{P}_n^+$ is an open subset of $\mathbb{R}^{n(n-1)}$. All statements about Lebesgue measure and absolute continuity below refer to these coordinates.

Theorem 5.1. *For Lebesgue-almost every $P \in \mathcal{P}_n^+$,*

$$K(P) = \{1\}. \quad (20)$$

More precisely, the exceptional matrices are contained in a finite union of zero sets of nonzero polynomials in the $n(n-1)$ free entries of P . Consequently, (20) holds almost surely under every probability law on $\mathcal{P}_n^+$ that is absolutely continuous in these coordinates.

Proof. The case $n = 1$ is immediate, so suppose $n \geq 2$. We first express the stationary probabilities as rational functions of the entries of P . Regard π_P^T as a column vector. In the system

$$(P^T - I)\pi_P^T = 0, \quad \sum_{j=1}^n \pi_P(j) = 1,$$

retain the first $n - 1$ equations from the first identity and add the normalization as the last equation. Let $M(P)$ be the resulting $n \times n$ coefficient matrix, whose last row consists of ones, and let e_n be the last coordinate vector, so that

$$M(P)\pi_P^T = e_n.$$

The discarded equation follows from the others, since the sum of the rows of $P^T - I$ is zero. The matrix $M(P)$ is invertible. Indeed, a nonzero vector in its kernel would have coordinate sum zero and would be stationary; adding a sufficiently small multiple of it to the strictly positive stationary probability vector would give a second stationary distribution, contrary to irreducibility.

Set $D(P) = \det M(P)$, and let $N_j(P)$ be the determinant obtained by replacing column j of $M(P)$ with e_n . Cramer's rule gives

$$\pi_P(j) = \frac{N_j(P)}{D(P)}, \quad D(P) \neq 0. \quad (21)$$

After the last entry of each row is eliminated, all of these determinants are polynomials in the free coordinates.

Fix a nonempty proper subset C and an integer r with $2 \leq r \leq n$. By (21), the equality $\pi_P(C) = 1/r$ is equivalent, on $\mathcal{P}_n^+$, to

$$Q_{C,r}(P) := r \sum_{j \in C} N_j(P) - D(P) = 0. \quad (22)$$

This polynomial is not identically zero. To see this, choose a strictly positive probability vector v such that $v(C) \neq 1/r$ and set $P = \mathbf{1}v$, with $\mathbf{1}$ now the column vector of n ones. Such a v exists because C and its complement are both nonempty. The stationary distribution of this matrix is v , so

$$Q_{C,r}(\mathbf{1}v) = D(\mathbf{1}v)(rv(C) - 1) \neq 0.$$

Recall that a nonzero polynomial in one variable has a Lebesgue-null zero set when it has finitely many roots. Inductively, write a nonzero polynomial in d variables as a polynomial in the last variable, with coefficients depending on the first $d - 1$ variables and choose one coefficient that is not identically zero. Outside of its null zero set, each one-dimensional section has finitely many roots. Application of Fubini's theorem over bounded boxes, then on their countable union, proves the corresponding assertion in dimension d .

It follows that each set in (22) is null. There are finitely many choices of C and r , and (19) places every exceptional matrix in their union. Outside that union, no consistent law can have more than one eventual class, and since a consistent law always exists, $K(P) = \{1\}$. Absolute continuity then yields the desired result. $\square$

Grimmett and Holmes [3, Remark 2.4 and Open Problem 3.8] use a particular law $\mathbb{Q}$ on transition matrices: The entries q_{ij} are independent and uniform on $(0, 1)$, and

$$P_{ij} = \frac{q_{ij}}{\sum_{k=1}^n q_{ik}}. \quad (23)$$

Theorem 5.1 gives the following answer to their question:

Corollary 5.2. *For the law $\mathbb{Q}$ in (23),*

$$\mathbb{Q}\{P : K(P) = \{1\}\} = 1.$$

Thus Open Problem 3.8 of [3] has an affirmative answer.

Proof. It suffices to check absolute continuity. For one row, suppress the row index and put $t = \sum_{j=1}^n q_j$ and $p_j = q_j/t$. The change of variables

$$q_j = tp_j \quad (1 \leq j \leq n), \quad p_n = 1 - \sum_{j=1}^{n-1} p_j$$

has absolute Jacobian t^{n-1} . The constraints $0 < q_j < 1$ become $0 < t < 1/\max_j p_j$, with p in the open probability simplex. Integrating out t gives the row density

$$g(p_1, \dots, p_{n-1}) = \int_0^{1/\max_j p_j} t^{n-1} dt = \frac{1}{n(\max_j p_j)^n}. \quad (24)$$

The rows are independent, so their joint law has the product of these densities. Theorem 5.1 applies, and for $n = 1$ the conclusion is immediate. $\square$

5.2 Matrices with identical rows

For the matrices used in the counterexample, the equal-mass partition condition is also sufficient. The following construction gives all possible numbers of classes and works for arbitrary positive real stationary weights.

Theorem 5.3. *Let π be a strictly positive probability vector on S , and let $P = \mathbf{1}\pi$. For $1 \leq r \leq n$,*

$$\begin{aligned} r \in K(P) &\iff S = C_1 \sqcup \dots \sqcup C_r, \\ &\pi(C_i) = 1/r \quad (1 \leq i \leq r) \text{ for some partition.} \end{aligned} \quad (25)$$

Moreover, every partition on the right can be realized as the eventual partition for every sequence of maps in the support of a consistent law.

Proof. Necessity follows from Theorem 4.2. For sufficiency, fix a partition $C_1, \dots, C_r$ with the stated masses. Choose a uniformly distributed permutation σ of $\{1, \dots, r\}$. Independently

of σ , choose independent representatives $Y_j \in C_j$ with

$$\mathbb{P}(Y_j = y) = r\pi_y, \quad y \in C_j; \quad r\pi(C_j) = 1 \quad (26)$$

and define the random map

$$F(x) = Y_{\sigma(i)}, \quad x \in C_i. \quad (27)$$

Use independent copies of these choices at successive times.

If $x \in C_i$ and $y \in C_j$, then

$$\mathbb{P}(F(x) = y) = \mathbb{P}(\sigma(i) = j)\mathbb{P}(Y_j = y) = \frac{1}{r} r\pi_y = \pi_y.$$

The law is therefore consistent with P . Every map collapses each C_i to a single point and maps different blocks into different blocks. The latter property persists under composition, since the induced actions on block labels are permutations. Hence all points of C_i meet at time one, while points from different initial blocks never meet. The eventual partition is exactly $\{C_1, \dots, C_r\}$ on every allowed sequence, as claimed. $\square$

There is a close antecedent in Gusev and Pribavkina [4, Theorem 8 and its proof]: Starting from a positive integer weight vector w , they use the directed multigraph with w_j edges from each vertex i to vertex j . Choosing an outgoing edge uniformly gives the identical-row transition matrix $P_{ij} = w_j / \sum_k w_k$. Their argument constructs a nonsynchronizing coloring when the weights admit a partition into at least two blocks of equal sum. Theorem 5.3 gives a direct random map formulation for each prescribed equal-mass partition, specifying its exact coalescence number and allowing arbitrary positive real weights.

In this terminology, a coloring is synchronizing if some finite composition of its maps is constant. For independent draws assigning positive probability to every map of the coloring, this is equivalent to $k(\mu) = 1$, by the minimum rank argument in Theorem 4.2. Thus the equal-weight obstruction used in synchronization theory is the same stationary mass obstruction used here.

5.3 The uniform matrix and the six-state matrix

Let P_n be the $n \times n$ matrix whose entries are all $1/n$.

Corollary 5.4. *For every $n \geq 1$,*

$$K(P_n) = \{r \in \{1, \dots, n\} : r \mid n\}. \quad (28)$$

Proof. The stationary distribution is uniform. If $r \in K(P_n)$, Corollary 4.3 gives $|C| = n/r$ for every eventual block, so r divides n . Conversely, if r divides n , partition S into r sets of cardinality n/r . Each has stationary mass $1/r$, and Theorem 5.3 supplies a consistent law with exactly r eventual classes. $\square$

The inclusion of every divisor in $K(P_n)$ already follows from Grimmett and Holmes [3, Theorem 5.5]. The reverse inclusion is a direct consequence of the classical equal-mass theorem. Together they settle the question stated at the beginning of [3, Section 6].

Corollary 5.5. *For the six-state matrix P in (3),*

$$K(P) = \{1, 2, 4\}. \quad (29)$$

Proof. Every subset has stationary mass equal to an integer multiple of $1/8$. If $r \in K(P)$, the mass $1/r$ of any eventual block must have this form. Thus r divides 8, and $r \leq 6$ leaves only 1, 2, 4.

All three values occur. A random constant map $x \mapsto J$, with J distributed according to π , gives one class. The coupling of Theorem 1.1 gives two. Finally, the four sets

$$\{2\}, \quad \{5\}, \quad \{1, 3\}, \quad \{4, 6\}$$

each have stationary mass $1/4$. Theorem 5.3 therefore gives a coupling with four classes. $\square$

The set in (29) describes the possible numbers of classes as the coupling law varies with P fixed. In Theorem 1.1, the law is fixed as well: it always gives two classes, but their cardinality profile varies. The two assertions concern different features of the same transition matrix.

6. Transformation semigroups and the preserved graph

The sixteen maps provide a small concrete instance of the transformation semigroup viewpoint developed in, for example, Budzban and Feinsilver [1]. The following facts can be verified directly:

Proposition 6.1. *The sixteen maps in (8) are closed under composition. Every such map has rank two. For any two of them, f and g ,*

$$\text{fib}(g \circ f) = \text{fib}(f), \quad (g \circ f)(S) = g(S). \quad (30)$$

Here $\text{fib}(f)$ denotes the partition into nonempty fibers of f .

Proof. Write $f = f_{b,u,v}$. Since $\{u, v\}$ is an edge and g preserves edges, the pair $(g(u), g(v))$ is an oriented edge. Hence

$$g \circ f_{b,u,v} = f_{b,g(u),g(v)}$$

is another map in the list. Because $g(u) \neq g(v)$, the two fibers of f remain distinct. Finally, g takes its two values on the endpoints of every edge, so $g(\{u, v\}) = g(S)$. $\square$

The semigroup therefore has exactly two fiber partitions, four image sets, and two bijections between the blocks of any chosen partition and the vertices of any chosen image, hence $2 \cdot 4 \cdot 2 = 16$ elements. Equation (30) explains why the rightmost, earliest map fixes the eventual classes while later maps change their locations.

There is also an exact graph interpretation. Given a transformation semigroup $\mathcal{T}$ on S , its *kernel graph* has an edge $\{x, y\}$ precisely when no $g \in \mathcal{T}$ identifies x and y ; see Schaefer [5]. In the present example the kernel graph is exactly G , not just a graph preserved by the maps. To see this, record the membership of each state in the two color classes:

x	1	2	3	4	5	6
$(\mathbf{1}_{A_0}(x), \mathbf{1}_{A_1}(x))$	(1, 1)	(0, 0)	(1, 1)	(1, 0)	(0, 1)	(1, 0).

Two states are separated by both partitions precisely when their bit pairs are complementary, which pairs are exactly the four edges in (5). Every other pair lies together in at least one of the two partitions and is identified by the corresponding maps.

Moreover, the sixteen maps are exactly the rank-two endomorphisms of G . If an edge-preserving map has rank two, its image must be an edge, and its two fibers give a proper two-coloring. Each of the two path components admits just two choices of color, so G has exactly two unordered proper bipartitions. Choosing the bipartition, an image edge, and the assignment of the two blocks to its endpoints gives exactly the sixteen maps above.

7. A direct connection to a previous coupling by state splitting

The preceding construction is not the only way to obtain varying cardinalities. A general state-splitting operation turns randomness of membership in a partition into randomness of ordinary cardinalities. We give its details to make the relationship to the earlier example explicit.

Let S be a finite state space with a random map coupling F_t . Replace each $x \in S$ by a nonempty finite set T_x , with the sets T_x pairwise disjoint, and let

$$T = \bigsqcup_{x \in S} T_x, \quad q : T \longrightarrow S, \quad q(y) = x \text{ for } y \in T_x.$$

Choose a strictly positive probability distribution η_x on each T_x . Independently at each time, choose a section $s_t : S \rightarrow T$ by choosing $s_t(x)$ according to η_x , independently over x and independently of F_t . The identity $q \circ s_t = \text{id}_S$ holds, and each section is injective. Define

$$\tilde{F}_t = s_t \circ F_t \circ q. \quad (31)$$

Proposition 7.1 (State splitting). *The lifted maps are independent and identically distributed. Their marginal transition matrix and stationary distribution are*

$$\tilde{P}_{yz} = P_{q(y), q(z)} \eta_{q(z)}(z), \quad \tilde{\pi}_z = \pi_{q(z)} \eta_{q(z)}(z). \quad (32)$$

If P is irreducible, so is $\tilde{P}$; if P is also aperiodic, so is $\tilde{P}$. The eventual lifted partition is

$$\tilde{\mathcal{C}}_\infty = \{q^{-1}(C) : C \in \mathcal{C}_\infty\}. \quad (33)$$

In particular, with $m_x = |T_x|$, its block cardinalities are the sums $\sum_{x \in C} m_x$ over the original eventual blocks.

Proof. The pairs (F_t, s_t) are independent and identically distributed. Conditioning on $F_t(q(y))$ proves the first formula in (32). Since $q \circ s_j$ is the identity, all intermediate sections cancel under composition:

$$\tilde{F}_t \circ \dots \circ \tilde{F}_1 = s_t \circ (F_t \circ \dots \circ F_1) \circ q \quad (t \geq 1). \quad (34)$$

Injectivity of s_t implies that the images of $y, z \in T$ on the left are equal exactly when $H_t(q(y)) = H_t(q(z))$. This proves (33); points in the same initial fiber of q meet at time one.

Summing the first formula in (32) over an intermediate fiber yields, by induction,

$$(\tilde{P}^t)_{yz} = (P^t)_{q(y), q(z)} \eta_{q(z)}(z), \quad t \geq 1. \quad (35)$$

Positive probabilities in the base chain therefore lift to positive probabilities between any chosen representatives. This proves irreducibility. The positive return times to y and to $q(y)$ also agree, proving the assertion about aperiodicity. Finally,

$$(\tilde{\pi} \tilde{P})_z = \sum_{x \in S} \pi_x P_{x, q(z)} \eta_{q(z)}(z) = \pi_{q(z)} \eta_{q(z)}(z) = \tilde{\pi}_z,$$

where the probabilities within every fiber sum to one. $\square$

This operation also preserves the stationary mass of each block:

$$\tilde{\pi}(q^{-1}(C)) = \pi(C). \quad (36)$$

The cardinality of the lifted block is $|q^{-1}(C)| = \sum_{x \in C} m_x$. Thus the choice of multiplicities determines how membership in the original blocks is reflected in their lifted sizes. The multiplicities are fixed before any maps are sampled, so variation in those sizes occurs under one fixed lifted coupling law. If distinct subsets of S have distinct sums of multiplicities,

each lifted block size identifies its original block. The entire lifted profile then determines the original partition, as used in the following corollary.

Corollary 7.2. *Any independent random map coupling of a finite irreducible chain whose eventual partition is not almost surely constant admits a finite state-splitting lift whose eventual cardinality profile is not almost surely constant.*

Proof. Enumerate $S = \{1, \dots, n\}$ and take $m_x = 2^{x-1}$. Distinct subsets of S have distinct sums of the m_x , by uniqueness of binary expansion. Thus the multiset of block sums uniquely determines the original unordered partition. Two distinct partitions of positive probability give two distinct lifted profiles of positive probability. Proposition 7.1 supplies the required irreducible coupling. $\square$

7.1 Splitting the four-state example

Grimmett and Holmes [3, Example 3.10] use the four maps

$$(1, 2, 1, 2), \quad (1, 2, 2, 1), \quad (3, 4, 3, 4), \quad (3, 4, 4, 3). \quad (37)$$

Their two possible eventual partitions are

$$\mathcal{D}_0 = \{\{1, 3\}, \{2, 4\}\}, \quad \mathcal{D}_1 = \{\{1, 4\}, \{2, 3\}\}. \quad (38)$$

Both have profile $(2, 2)$. Take equal probabilities on the four maps. Directly from (37), the transition matrix is

$$P_0 = \begin{pmatrix} 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix}. \quad (39)$$

Note that P is irreducible, aperiodic and doubly stochastic. Each map has image either $\{1, 2\}$ or $\{3, 4\}$, and every map separates both of these pairs. Thus the first map fixes the eventual partition, and the two possibilities in (38) each have probability $1/2$.

Now choose multiplicities

$$(m_1, m_2, m_3, m_4) = (2, 1, 2, 1), \quad (40)$$

with uniform section choices in each fiber. There are six lifted states. The block sums are

base partition	lifted block sizes	sorted profile
$\mathcal{D}_0$	$(m_1 + m_3, m_2 + m_4) = (4, 2)$	$(2, 4)$
$\mathcal{D}_1$	$(m_1 + m_4, m_2 + m_3) = (3, 3)$	$(3, 3)$

Proposition 7.1 gives a second six-state counterexample, again with probability $1/2$ for each profile. Its stationary masses are $1/8$ at the four copies of states 1 and 3 and $1/4$ at the copies of states 2 and 4.

This lift has eight distinct maps with equal probabilities: For each base map, its image contains one doubled state and one single state, and only the section choice at that doubled image state affects the lifted map. Note that this is a different coupling from the sixteen-map construction. In particular, its transition matrix retains zeros from (39); it is irreducible and aperiodic but not strictly positive. The lift proves that an elementary modification of the earlier random partition example suffices to produce random cardinalities.

Appendix. The complete map list and exact verification

For each oriented edge (u, v) , the two maps in the corresponding row below are $f_{0,u,v}$ and $f_{1,u,v}$. Every entry is the ordered tuple $(f(1), \dots, f(6))$. Each of the sixteen maps has probability $1/16$.

(u, v)	$f_{0,u,v}$	$f_{1,u,v}$
(1, 2)	(1, 2, 1, 1, 2, 1)	(1, 2, 1, 2, 1, 2)
(2, 1)	(2, 1, 2, 2, 1, 2)	(2, 1, 2, 1, 2, 1)
(2, 3)	(2, 3, 2, 2, 3, 2)	(2, 3, 2, 3, 2, 3)
(3, 2)	(3, 2, 3, 3, 2, 3)	(3, 2, 3, 2, 3, 2)
(4, 5)	(4, 5, 4, 4, 5, 4)	(4, 5, 4, 5, 4, 5)
(5, 4)	(5, 4, 5, 5, 4, 5)	(5, 4, 5, 4, 5, 4)
(5, 6)	(5, 6, 5, 5, 6, 5)	(5, 6, 5, 6, 5, 6)
(6, 5)	(6, 5, 6, 6, 5, 6)	(6, 5, 6, 5, 6, 5)

The accompanying source package contains a short script, `verify.py` which checks the marginal transition matrix, edge preservation, closure under all 16^2 compositions, and the eight-map state-splitting construction.

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